

SHALKAR, Kazakhstan

WMO: 356330

Lat: 47.850N

Lon: 59.617E

Elev: 176

StdP: 99.23

Time Zone: 5.00 (E05)

Period: 84-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.6	-26.1	-31.3	0.2	-28.1	-29.0	0.3	-25.8	16.8	-11.2	15.4	-12.7	3.0	270	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	35.8	18.5	34.1	17.9	32.3	17.4	19.9	30.4	19.2	30.4	18.5	29.7	4.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.3	22.3	15.5	11.3	22.2	14.2	10.3	22.1	57.5	30.0	55.1	30.7	52.8	29.7	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	12.0	10.3	-31.7	39.1	1.5	1.3	-32.7	40.0	-33.6	40.7	-34.4	41.4	-35.5	42.3		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-12.3	-14.4	-7.5	8.1	15.6	22.8	24.5	22.4	15.5	6.3	-3.7	-9.8
	DBStd	15.01	7.09	6.46	6.99	5.95	4.42	4.16	3.33	3.59	4.49	4.86	6.26	6.55
	HDD10.0	3200	691	684	544	103	8	0	0	0	9	136	412	613
	HDD18.3	5117	949	917	802	310	104	10	1	9	106	374	662	872
	CDD10.0	1636	0	0	0	45	183	383	448	383	173	21	0	0
	CDD18.3	509	0	0	0	2	21	143	191	133	20	0	0	0
	CDH23.3	6556	0	0	0	28	323	1933	2415	1546	299	11	0	0
	CDH26.7	2962	0	0	0	4	81	945	1186	666	78	2	0	0
Wind	WSAvg	4.7	4.8	5.2	5.4	5.2	4.8	4.5	4.4	4.1	4.3	4.2	4.5	4.2
Precipitation	PrecAvg	185	16	12	15	19	13	15	14	13	12	18	18	16
	PrecMax	421	39	42	45	172	56	78	52	46	73	54	53	36
	PrecMin	89	0	0	1	0	0	0	0	0	0	2	1	0
	PrecStd	66	11	10	11	30	13	16	15	14	15	13	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.0	12.4	26.4	31.6	38.4	38.9	36.1	30.9	24.4	9.9	2.4
		MCWB	0.5	0.1	7.6	14.3	15.5	19.1	20.6	18.1	14.9	12.7	7.2	0.8
	2%	DB	0.4	0.0	7.1	23.2	28.6	36.1	35.8	34.0	28.9	21.0	6.5	1.1
		MCWB	-0.1	-0.5	4.2	12.6	14.8	18.3	18.8	17.5	14.6	10.9	3.3	0.3
	5%	DB	-0.9	-2.0	4.0	20.4	26.6	33.7	34.3	32.0	26.8	17.6	5.2	0.1
		MCWB	-1.5	-2.6	2.0	11.3	14.2	17.7	18.3	17.3	14.0	9.1	2.9	-0.7
	10%	DB	-2.3	-4.3	1.7	17.9	24.8	31.7	32.6	30.3	24.6	14.8	3.7	-0.9
		MCWB	-2.9	-4.8	0.4	9.7	13.5	16.9	17.9	16.8	13.2	8.1	1.6	-1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.4	7.8	15.2	17.5	20.4	23.6	19.7	18.0	13.6	7.8	1.3
		MCDB	0.8	0.7	12.3	23.0	25.5	31.4	34.3	30.1	24.7	23.5	9.0	1.7
	2%	WB	0.0	-0.5	4.4	13.6	16.2	19.5	20.5	19.0	16.4	11.4	4.6	0.4
		MCDB	0.3	0.0	6.7	22.1	25.2	31.8	30.6	29.3	23.7	18.8	5.6	0.9
	5%	WB	-1.4	-2.5	2.1	11.9	15.2	18.7	19.5	18.4	15.2	10.0	3.2	-0.5
		MCDB	-0.9	-1.9	3.8	19.2	24.0	31.7	30.1	28.6	24.4	16.4	4.6	0.1
	10%	WB	-2.9	-4.7	0.5	10.3	14.4	17.8	18.8	17.7	14.0	8.4	1.7	-1.8
		MCDB	-2.4	-4.3	1.6	17.1	22.9	29.9	29.9	27.5	23.7	14.0	3.2	-1.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.3	8.9	11.4	13.1	13.9	13.2	13.3	14.0	12.5	8.0	7.6
		MCDBR	7.4	7.1	9.3	14.5	16.2	16.4	15.2	15.4	17.3	17.2	9.9	6.9
		MCWBR	7.3	6.8	6.7	7.3	6.7	5.6	5.0	5.8	7.8	9.4	7.8	6.1
	5% WB	MCDBR	7.5	6.9	8.8	14.3	13.9	14.5	13.1	13.3	14.8	15.5	8.6	6.2
		MCWBR	7.5	6.8	6.5	7.7	6.4	5.8	5.1	5.4	7.3	9.1	7.4	5.7
Clear-Sky Solar Irradiance	taub	0.308	0.334	0.402	0.438	0.436	0.430	0.419	0.417	0.365	0.360	0.340	0.317	
	taud	2.100	2.022	2.018	2.037	2.118	2.197	2.247	2.233	2.330	2.257	2.221	2.133	
	Ebn at Noon	729	794	796	809	829	836	840	824	837	768	692	663	
	Edh at Noon	89	121	145	157	150	140	131	128	105	96	80	77	
All-Sky Solar Radiation	RadAvg	1.47	2.60	3.69	5.00	6.42	6.95	6.73	5.95	4.60	2.76	1.41	0.98	
	RadStd	0.25	0.36	0.35	0.49	0.44	0.42	0.32	0.32	0.23	0.26	0.15	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page